

MSE 360: Kinetic Processes in Materials

Homework 3

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. Answer true or false (and why):
 - i. For the reaction $A + 2B \rightarrow C$, the rate equation must be: $\frac{dC_A}{dt} = -kC_AC_B^2$.
 - ii. Decreasing the activation energy of a reaction by half will increase the reaction rate by a factor of 2.
2. Consider the integrated second-order rate law for a reaction which is first order with respect to two distinct reactants (A and B). Prove that the following expressions (Eqns. 3.52 and 3.53 in the textbook) are equivalent:

$$C_A = \frac{C_{B0} - \frac{b}{a}C_{A0}}{\frac{C_{B0}}{C_{A0}} \exp\left[-kt\left(\frac{b}{a}C_{A0} - C_{B0}\right)\right] - \frac{b}{a}}; \frac{C_A}{C_B} = \frac{C_{A0}}{C_{B0}} \exp\left[kt\left(\frac{b}{a}C_{A0} - C_{B0}\right)\right]$$

3. If the rate constant k for a first-order reaction doubles with a change in temperature from $T = 300 \text{ K}$ to $T = 320 \text{ K}$, what is ΔG_{act} ?
4. For a 0, 1st, and 2nd-order rate reaction with respect to the concentration of reactant A, explain how to determine k , the rate constant, from data of the concentration of A vs. time. *Use graphs to support your explanation.*